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Infections due to contamination of products used in healthcare settings

This report briefly summarises UKHSA's recent experience of investigating and responding to incidents involving contamination of products used in healthcare settings. It identifies an emerging patient safety concern and provides information and recommendations for healthcare professionals to reduce risks.

Context

Contamination of products used in healthcare settings is an uncommon but emerging patient safety concern globally. UKHSA has identified and responded to several incidents involving contamination of products used in UK healthcare settings in recent years. Products affected have included items intended for cosmetic and hygiene purposes, as well as a small number of medical devices and medicines. Contamination of such products has the potential to expose large numbers of people across a wide geography. Micro-organisms involved are typically opportunistic pathogens which do not usually affect healthy people. However, they may cause infections in some individuals, particularly those with underlying health conditions or requiring complex medical care.

Examples of recent incidents, products affected and impact

UKHSA has worked with UK and international partners to investigate clusters of infections suspected or proven to be associated with contaminated products.

The table below provides a summary of 7 recent incidents where a contaminated product was identified as the source of infections and which required UKHSA response in partnership with the NHS, regulators and health system stakeholders. Products linked to infections have mostly been non-sterile, water-based products. These have included ultrasound gels, skin cleansing wipes, patient and surface cleaning wipes, and lubricating eye gels. For some incidents it has not been possible to identify the source.

Good clinical and infection prevention and control (IPC) practice plays an important role in preventing and mitigating risk. This includes the use of sterile products for invasive procedures and for ongoing care of intravenous

lines and wounds. Opportunistic pathogens causing contamination are commonly found in the environment, such as in soil and water, and rarely cause infections in healthy people. However, they can cause severe infections in people with weakened immune systems or when they are introduced into body sites such as the bloodstream or respiratory tract.

Recent incidents have resulted in serious presentations including bloodstream infections and pneumonia. One patient in England is known to have died as a direct consequence. Infections have particularly, though not exclusively, affected patients in critical care settings and those with intravenous lines in place. Those with central venous lines used to administer therapy over a prolonged period appeared to be at higher risk of developing bloodstream infections. Patients with certain medical conditions are at increased risk from particular opportunistic pathogens; for example, individuals with cystic fibrosis (CF) are at higher risk of developing severe infections from Burkholderia cepacia complex (Bcc).

Challenges and control measures

There are multiple challenges to identifying and responding to contamination incidents; these include:

- under-detection – microorganisms causing such incidents rarely cause infections and therefore may not be detected or considered significant by clinicians and laboratories
- challenges in identification – there are many products that could be affected which can be obtained over the counter or bought online, are generally not recorded by healthcare staff, may be used by patients outside of healthcare settings, or brought in by patients or visitors
- testing limitations – contamination may be low level and variable across batches or involve multiple microorganisms, making it difficult to identify and/or link to infections
- challenges in removing contaminated products from use – for example, product recalls are not always possible, timely or completely effective

Control measures and communications have included: product recalls; issuing of product safety notices; NHS Supply Chain customer notices (where products have been distributed via this route); UKHSA Briefing Notes to the health system (for all incidents); and provision of IPC information and guidance to healthcare professionals, patients and the public. Due to risk of patient harm, UKHSA has also issued National Patient Safety Alerts for several incidents; these have contained information and advice, including reinforcing IPC guidance for intravenous line care (such as avoiding use of non-sterile wipes) and use of sterile wipes for wound care.

Although such incidents remain rare, UKHSA continues to work with the NHS and partners to prevent and respond to occurrences.

Recommendations for healthcare professionals

It is recommended that clinical, laboratory and IPC teams support the identification of relevant pathogens and clusters of infections. Product contamination incidents may affect individuals over a prolonged time period and across a wide geography, including multiple healthcare facilities.

To support investigations, clinicians should:

- consider a potential link to contaminated products when opportunistic/water-borne pathogens (for example, Burkholderia spp) are detected in clinical specimens (even a single specimen), especially if these are from patients at increased risk, including those in critical care settings and/or those with longstanding intravenous lines
- notify their local UKHSA Health Protection Team (https://www.gov.uk/health-protection-team) promptly if a cluster of related infections is identified or an outbreak suspected
- retain the relevant product(s) if contamination is suspected, as testing may be required as part of the investigation process

Additionally, NHS and independent laboratories should:

- submit isolates from any new uncommon Gram-negative bacteria (such as Burkholderia spp., Ralstonia spp., Pandoraea spp.) bloodstream infection, including any new isolates from CF patients to the UKHSA Antimicrobial Resistance and Healthcare Acquired Infections reference laboratory (see Bacteriology reference department manual (https://www.gov.uk/government/publications/bacteriology-reference-department-brd-user-manual) for further information)
- assist UKHSA with incident investigation and response if requested to do so

IPC and clinical teams managing patients with longstanding intravenous lines or critical care patients should consider relevance of this information for local policy and guidance, and in providing advice for patients. For example, reinforcing the need to ensure only sterile wipes are used for intravenous line care. Sterile products are usually clearly marked as ‘sterile’ on the packaging.

IPC, clinical and procurement leads should consider the risk associated with the use of non-sterile products (such as ultrasound gels, wipes, creams or

ointments) in critical care units and where patients with intravenous lines and at higher risk of opportunistic infection are being managed.

It is recommended that procurement leads order products for the NHS through NHS Supply Chain. NHS Supply Chain applies robust procurement standards and governance and can support interventions to mitigate risk when contamination incidents are identified.

Summary of 7 recent product contamination incidents responded to by UKHSA 2010 to 2025

Product affected (date of incident) and communications issued	Contaminating organism	Confirmed cases	Description of patients affected
Ultrasound gel (2010 to 2022) Communications included: patient safety alert, IPC guidance (1) and journal publication (2)	Burkholderia cepacia ST767	153	• cases reported by 67 UK hospitals • invasive isolates recovered from 114 (75%) cases • 41 cases (36% of 118 with information available) managed in critical care settings at time of sample
Environmental cleaning wipes (2020 to 2022) Communications included: product recall and journal publication (2)	Burkholderia contaminans ST1891	66	• cases identified from 42 UK hospitals • invasive isolates recovered from 41 (72%) cases • 3 (6% of 49 with information) managed in critical care settings at time of sample • 33 (75% of 44 with information) had recent intravenous access device insertion or removal • 8 (32% of 25 with information) were

Product affected (date of incident) and communications issued	Contaminating organism	Confirmed cases	Description of patients affected
			considered immunocompromised
Non-sterile alcohol-free skin cleansing wipes for first aid use (2020 to 2025) Communications included: patient safety alert and GOV.UK news story	Burkholderia stabilis ST480	57	• cases reported from 42 UK hospitals • invasive isolates recovered from 41 (72%) cases • 3 (6% of 49 with information) managed in critical care settings at time of sample • 33 (75% of 44 with information) had recent intravenous access device insertion or removal • 8 (32% of 25 with information) were considered immunocompromised
Patient hygiene wet wipes and other hygiene products (2021 to 2022) Communications included: patient safety alert, product recall and journal publication (3)	Pseudomonas aeruginosa ST3875	12	• cases identified from 2 UK hospitals • invasive isolates recovered from 2 (17%) cases; • 10 (100% of 10 with information) managed in critical care settings at time of sample • over 400 cases in Norway (38% ICU patients)
Saline products for irrigation, inhalation and eye wash (2023 to 2024) Communications	Ralstonia pickettii	4	• cases identified from 4 UK hospitals • invasive isolates recovered from 4 (100%) cases • no cases reported from critical care

Product affected (date of incident) and communications issued	Contaminating organism	Confirmed cases	Description of patients affected
included: product recall and journal publication (4)			settings • 4 (100%) had a recent intravenous access device insertion or removal • 4 (100%) were immunocompromised
Carbomer eye gel (2023 to 2024) Communications included: patient safety alert, product recall, device safety information and HPR (5)	Burkholderia cepacia ST2147	66	• cases identified from 29 UK hospitals • invasive isolates recovered from 8 (14%) cases • 31 (58% of 53 with information) managed in from critical care settings at time of sample • 48 (94% of 51 with information) had significant comorbid conditions including 2 with cystic fibrosis • 1 attributable death
Ultrasound gel (2024) Communications included: product recall, revised IPC guidance (6) and HPR (7)	Burkholderia stabilis ST1565	7	• cases identified from 5 UK hospitals • invasive isolates recovered from 7 (100%) cases • 3 (43%) managed in critical care settings at time of sample • 5 (83% of 6 with information) had a recent intravenous access device insertion or removal

Notes: “Invasive isolate” denotes that the sample was recovered from a normally sterile site of the body, for example blood or deep tissue. “Critical

care" includes high acuity settings such as Intensive Care Units (ICU), neonatal ICU, High Dependency Units, etc. "Immunocompromised" was defined by clinical teams managing cases.

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Update on lead exposure in children in England: 2024 cases

Lead is a heavy metal which is a persistent environmental pollutant and exposure to lead adversely affects multiple body systems. There is no known safe threshold for lead exposure. Lead is particularly harmful to young children and pregnant women (accumulated lead stored in bones is mobilised into the bloodstream during pregnancy thus exposing the developing foetus). The harmful health effects of lead exposure are completely preventable (1).

The UKHSA has coordinated the national Lead Exposure in Children Surveillance System (LEICSS) since 2016; following evaluation of a successful pilot study (Lead Poisoning in Children, LPIC). LEICSS collects data from participating laboratories (the Supra Regional Assay laboratories and some participating hospital laboratories). LEICSS is now integrated into the UKHSA Lead Exposure Public Health Intervention and Surveillance (LEPHIS) group.

The eighth annual report of LEICSS data has now been published (2). It summarises all children notified to the surveillance system from 1 January to 31 December 2024, aged 0 to 15 years, with an elevated blood lead concentration of ≥0.24µmol/L (≥5µg/dl), who require public health intervention to identify and eliminate the sources of exposure.

A total of 247 cases were notified in 2024; the key findings of the report are that:

- most cases (78%) were directly notified through the surveillance system by participating laboratories; 10% were notified through other routes (for example, clinicians reporting directly to health protection teams) which is similar to previous years
- the average detection rate for England in 2024 was 23 cases per million children, although there was significant regional variation
- cases are typically 1 to 4 years of age (64%), male (58%), and more likely to be resident in the most deprived areas of England (48%), compared with 24% of the overall population of children aged 0 to 15 years who live in these areas
- the most common reported exposures in 2024 (101 completed Enhanced Surveillance Questionnaires (ESQ) were soil (24%) and paint (17%). The majority of cases exhibited pica behaviour (94%) and many also had learning difficulties (74%) (see note 1)

- the number of cases reported to LEICSS is significantly lower than the expected incidence of lead exposure in children in England, based on international population surveys (see note 2) (3-8)

The main messages and recommendations include that:

- there is no safe level of exposure to lead and children who exhibit pica or other hand-to-mouth behaviour in areas with lead hazards face the greatest risk of exposure
- clinicians should be aware of the risk of lead exposure for children, the common sources of exposure, children most likely to be at risk, presenting symptoms and signs of exposure
- children meeting the case definition (aged under 16 years, BLC ≥5µg/dL or ≥0.24µmol/L) must be reported to UKHSA health protection teams for active public health case management

Notes

1. Case management requires cases with a BLC ≥0.48 µmol/L (equivalent to ≥10µg/dL) to complete an ESQ; therefore, not all cases are expected to have an ESQ completed.
2. It is estimated that between 34,541 (using US estimate that 1.1% of children aged 1 to 5 years have a BLC ≥ 0.24 µmol/L , applied to UK children aged 1 to 5 years) and 57,058 (using France estimate that 1.5% of children aged 1to 6 years had a BLC ≥ 0.24 µmol/L, applied to UK children aged 1 to 6 years) children in the respective age groups in the UK might be expected to have BLCs above the intervention concentration.

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Surveillance of surgical site infections in NHS hospitals in England: 2024 to 2025 annual report

The latest surgical site infections (SSI) annual report has been published, summarising data submitted to the UKHSA national Surgical Site Infection Surveillance Service by 190 NHS hospitals and 8 independent sector NHS treatment centres in England. The report presents SSI risk benchmarks for each of 17 surgical categories, trends in SSI incidence, variation between participating hospitals and risk-stratified SSI incidence.

Surveillance data on 146,411 operations and 1,250 surgical site infections (SSIs) detected during inpatient stay or on readmission to hospital were submitted in 2024 to 2025 (1). The number of procedures submitted increased by 7.9% compared to the previous year, and by 15.3% compared to pre-pandemic year (2019 to 2020). NHS trusts performing surgery in any of 4 orthopaedic categories (hip replacement, knee replacement, reduction of long bone fracture and repair of neck of femur) are required to undertake surveillance of SSI for a minimum of one 3-month surveillance period per financial year. Trust-level SSI risk results for mandatory orthopaedic categories can be found as accompanying supplementary tables (2).

Key findings include:

- all except two eligible trusts met the mandatory participation surveillance requirements in 2024 to 2025
- 5 trusts were identified as high outliers in for orthopaedic surgery, an increase from 2 in the previous year
- 10-year trends in SSI risk varied by surgical category, with 2 out of 10 categories (knee replacement and spinal surgery) seeing overall decreases in annual inpatient and readmission SSI risk and the remaining 8 categories showing stable trends during this period
- there was an indication of elevated SSI risk among Asian patients undergoing large bowel surgery (unadjusted Risk Ratio (uRR) 1.55, 95% CI: 1.04 to 2.33) and reduced SSI risk in Asian patients undergoing coronary artery bypass graft surgery (uRR 0.62; 0.50 to 0.77) compared to White patients
- among SSIs with accompanying microbiology data, Enterobacterales continued to make up the largest proportion of isolates for both superficial SSIs (32.0%) and deep incisional or organ and space SSIs (28.0%)
- the proportion of SSIs that were deep incisional increased in 2024/2025 in patients undergoing reduction of long bone fracture (50.0% to 70.0%) and spinal surgery (48.6% to 62.1%) compared to 2023/2024
- the median time to infection was 9 days for procedures without an implant and 19 days for procedures including an implant

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Infection reports in this issue

Laboratory confirmed cases of invasive meningococcal infection in England: July to September 2025 (https://www.gov.uk/government/publications/laboratory-confirmed-cases-of-imd-in-england-july-to-september-2025)

Decreases in chlamydia tests, diagnoses and positivity in young people, England 2023 to 2024 (https://www.gov.uk/government/publications/chlamydia-analyses-to-inform-control-interventions)

Tetanus in England: 2024 (https://www.gov.uk/government/publications/tetanus-in-england-annual-reports)

Campylobacter and non-typhoidal Salmonella infections in England: Q1 2023 to Q3 2025 (https://www.gov.uk/government/publications/gastrointestinal-infections-in-england)

Vaccine coverage

Prenatal pertussis vaccination coverage in England from July to September 2025 (https://www.gov.uk/government/publications/pertussis-immunisation-in-pregnancy-vaccine-coverage-estimates-in-england-october-2013-to-march-2014/prenatal-pertussis-vaccination-coverage-in-england-from-july-to-september-2025)

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Chemical and environmental hazards

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